

ORAI1 Deficiency (CRAC Channelopathy)

— Disease Characterization Report

Autonomous discovery report. Evidence tiers indicated as: [Human clinical], [Model organism], [In vitro], [Computational/inferred]. Primary citations given as PMID.

Summary / Consensus Answer

ORAI1 Deficiency is an ultra-rare, autosomal-recessive primary immunodeficiency caused by **loss-of-function mutations in *ORAI1***, the gene encoding the pore-forming subunit of the plasma-membrane **Ca²⁺ release-activated Ca²⁺ (CRAC) channel**. Loss of ORAI1 abolishes **store-operated Ca²⁺ entry (SOCE)**, crippling the Ca²⁺/calcineurin/NFAT signaling axis. The disease — part of the broader entity "**CRAC channelopathy**" (which also includes STIM1 loss-of-function) — presents as a clinical tetrad: (1) a **severe-combined-immunodeficiency (SCID)-like combined immunodeficiency** with recurrent/severe infections, (2) **autoimmunity/lymphoproliferation**, (3) **non-progressive congenital muscular hypotonia (myopathy)**, and (4) **anhidrotic ectodermal dysplasia** with defective sweating and **dental enamel defects (amelogenesis imperfecta)**. Hematopoietic stem cell transplantation (HSCT) can cure the immunological disease but does not correct the non-hematopoietic (sweat gland, dental, muscle) manifestations.

Important nomenclature note: *ORAI1* is a "two-faced" gene. **Loss-of-function** mutations (the subject of this report) cause the recessive immunodeficiency/CRAC channelopathy described here. **Gain-of-function** mutations in the same gene cause dominant **tubular aggregate myopathy (TAM)** and **Stormorken syndrome** — these are distinct diseases and are NOT "ORAI1 deficiency." They are mentioned only for contrast (

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1. Disease Information

- **Overview:** A monogenic (Mendelian) inborn error of immunity in which absent CRAC-channel function eliminates SOCE in immune and non-immune cells. Because SOCE is the dominant Ca^{2+} -influx pathway in T lymphocytes, sweat glands, and tooth-forming ameloblasts, the disorder is multisystem.
 - **Key identifiers:**
 - **OMIM (phenotype):** Immunodeficiency 9 (IMD9) #612782 ("combined immunodeficiency with anhidrotic ectodermal dysplasia").
 - **OMIM (gene):** *ORAI1* #610277.
 - **Orphanet:** "CRAC channelopathy" / "Immunodeficiency due to a defect in CRAC channel" (Orphanet groups *ORAI1* and *STIM1* LOF together).
 - **MonDO:** "immunodeficiency 9" / "combined immunodeficiency due to *ORAI1* deficiency" (maps to OMIM #612782; a subtype under immunodeficiency-with-ectodermal-dysplasia); umbrella entity "CRAC channelopathy." (Exact MONDO accession should be verified against the current MONDO release; the OMIM #612782 cross-reference is the stable anchor.)
 - **ICD-10:** D81.8 / D81.9 (combined immunodeficiencies, other/unspecified). **ICD-11:** 4A01.1Y (combined immunodeficiencies).
 - **MeSH:** *ORAI1* Protein (D000071140); Severe Combined Immunodeficiency (D016511) as related.
 - **HGNC:** *ORAI1* HGNC:25896; **UniProt** Q96D31; **Ensembl** ENSG00000276045 / ENSG00000152580.
 - **Synonyms / alternative names:** Immunodeficiency 9; CRAC channelopathy (*ORAI1*-type); Combined immunodeficiency due to *ORAI1* deficiency; *ORAI1*-CID; "Anhidrotic ectodermal dysplasia with immunodeficiency" (a distinct, *ORAI1*-specific form — noted by

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 as distinct from NEMO/IKBKG-related EDA-ID); SCID-like disease with SOCE defect. Older literature: "CRAC channel deficiency."
 - **Data source type:** Information is **disease-level and individual-patient (case-report/case-series)** derived. Because <20–30 patients are reported worldwide, evidence comes from small kindreds, functional in-vitro studies of patient cells, and mouse models — not from population EHR or registry aggregates.
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2. Etiology

Disease Causal Factors

- **Primary cause = genetic.** Biallelic (homozygous or compound heterozygous) loss-of-function variants in *ORAI1* that abolish CRAC-channel function and SOCE [Human clinical;

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- No environmental, infectious, or toxic cause initiates the disease; environmental agents act only as **downstream triggers of complications** (opportunistic infections due to the immunodeficiency).

Risk Factors

- **Genetic:** The causal biallelic *ORAI1* LOF genotype is necessary and sufficient. **Consanguinity** is a major risk factor because homozygous null alleles predominate in reported kindreds (recessive rare disease). Family history of an affected sib or of consanguineous union raises risk.
- **Environmental / demographic:** None established as causal. **Heat exposure** is a hazard (not a cause) because anhidrosis predisposes to hyperthermia/heat intolerance. Sex is not a risk factor (autosomal).

Protective Factors

- **Genetic:** Heterozygous carriers are clinically unaffected (recessive), although their T cells show partially reduced SOCE — a **gene-dosage effect** — demonstrating that ~50% channel function is protective [In vitro;

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Redundant Ca^{2+} -entry contributions from paralogs **ORAI2/ORAI3** and **STIM2** may partially buffer some cell types (basis of variable non-immune expressivity) [Model organism/in vitro;

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- **Environmental:** Avoidance of heat, infection prophylaxis, and immune reconstitution are protective interventions (see §12–13), not innate protective factors.

Gene–Environment Interactions

- The genetic immunodeficiency sets susceptibility; **pathogen exposure** determines which infections manifest. Autoimmune flares may be triggered by infection due to loss of regulatory T-cell tolerance. No formal GxE quantitative data exist for this ultra-rare disease.
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3. Phenotypes

Frequencies are qualitative (small n). HPO term suggestions included.

Phenotype	Type	Onset	Severity / Course	Frequency	HPO
Recurrent/severe/opportunistic infections (viral, bacterial, fungal; CMV, EBV, Pneumocystis, candidiasis)	Clinical sign	Neonatal–infantile	Severe, life-threatening; often the presenting feature	Nearly all	HP:0002719 (Recurrent infections), HP:0004430 (Severe combined immunodeficiency)
Combined immunodeficiency with normal lymphocyte counts but impaired T-cell function	Lab abnormality	Congenital	Persistent	~All	HP:0005387 (T-cell dysfunction), HP:0002090-adjacent
Autoimmunity: autoimmune hemolytic anemia, autoimmune thrombocytopenia, lymphoproliferation/HLH	Clinical sign/lab	Infancy–childhood	Variable, can be severe (HLH reported as presenting event)	Common	HP:0001973 (Autoimmune thrombocytopenia), HP:0001890 (Autoimmune hemolytic anemia), HP:0002665 (Lymphadenopathy)
Non-progressive congenital muscular hypotonia / myopathy	Physical/sign	Congenital/neonatal	Non-progressive, generally mild-moderate	Most	HP:0008947 (Infantile muscular hypotonia), HP:0003198 (Myopathy)
Anhidrosis / hypohidrosis (defective sweating) → heat intolerance	Physical sign	Congenital	Lifelong, stable	Most	HP:0000970 (Abnormality of the sweat glands), HP:0000966 (Hypohidrosis), HP:0009927 (Anhidrosis)
				Most	

Phenotype	Type	Onset	Severity / Course	Frequency	HPO
Amelogenesis imperfecta / dental enamel defects	Physical sign	With tooth eruption (childhood)	Lifelong, stable		HP:0000705 (Amelogenesis imperfecta), HP:0006297 (Hypoplastic enamel)
Ectodermal dysplasia (broader)	Physical	Congenital	Stable	Common	HP:0000968 (Ectodermal dysplasia)
Failure to thrive / growth failure (secondary to infection)	Sign	Infancy	Variable	Common	HP:0001508 (Failure to thrive)
Splenomegaly/ hepatomegaly (lymphoproliferation)	Sign	Childhood	Variable	Subset	HP:0001744 (Splenomegaly)

- **Quality-of-life impact:** High burden — recurrent hospitalizations for infection; lifelong infection prophylaxis; heat intolerance restricts activity/climate tolerance; dental restoration needs; muscle weakness affects motor milestones. Without immune reconstitution, early mortality is common. No formal EQ-5D/SF-36 data exist (ultra-rare).

- **Evidence:**

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(subset deficits, enamel, anhidrosis, muscular dysplasia);

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(tetrad);

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(HLH as presenting manifestation of profound CID).

4. Genetic / Molecular Information

- **Causal gene:** *ORAI1* (calcium release-activated calcium modulator 1), **12q24.31**, HGNC:25896, OMIM gene #610277, UniProt Q96D31. Encodes a 301-aa tetraspanning (4-transmembrane) plasma-membrane protein; **six ORAI1 subunits assemble into the hexameric CRAC-channel pore** (concatemer functional analysis,

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Drosophila Orai hexamer structure PDB 4HKR). Key pore/structural residues: **E106** (Ca²⁺ selectivity filter, TM1), **R91** (TM1); disease missense mutations (R91W, G98R, L194P) map to transmembrane/pore-proximal regions.

- **Pathogenic variants (all LOF, biallelic; germline):**
- **c.271C>T, p.Arg91Trp (R91W)** — first reported, homozygous; misfolds/mis-gates channel, abolishes CRAC current. Identified by combined SNP-array linkage + a *Drosophila* RNAi screen; WT ORAI1 re-expression rescued SOCE/I_CRAC in patient T cells, proving causality [PMID: 16582901].
- **p.Val181SerfsX8 (frameshift), p.Leu194Pro (missense), p.Gly98Arg (missense)** — homozygous; suppress ORAI1 protein expression and SOCE [PMID: 29155098].
- Additional reported nulls include a nonsense/premature-stop allele (A88Sfs-type) abrogating protein.
- **Variant classes:** missense (pore/TM), frameshift, nonsense/premature termination → all functionally **loss-of-function** (no/absent channel or non-conducting channel).
- **ACMG classification:** reported alleles are **Pathogenic/Likely Pathogenic** (functional SOCE assays = strong PS3 evidence; recessive segregation; absent SOCE).
- **Allele frequency:** Causal alleles are **absent or ultrarare in gnomAD**; *ORAI1* is relatively tolerant to heterozygous LOF but biallelic LOF is nearly private to consanguineous kindreds.
- **Somatic vs germline:** **Germline** exclusively.
- **Functional consequence: Loss of function** (recessive). Contrast: **gain-of-function** ORAI1 (e.g., p.G98S, p.L138F, p.P245L, p.T184M) → constitutive SOCE → TAM/Stormorken (dominant) — different disease [PMID: 26469693].
- **Modifier genes:** Paralogous **ORAI2, ORAI3** and ER sensors **STIM1, STIM2** contribute to residual/tissue-specific SOCE and likely modify expressivity of non-immune features [PMID: 28294127].

- **Epigenetic information:** No disease-specific methylation/histone signature reported; not implicated.
- **Chromosomal abnormalities:** None — disease is due to point/small variants, not CNVs or aneuploidy.

5. Environmental Information

- **Environmental factors:** None causal. Pathogen exposure is the principal external factor determining infectious complications.
- **Lifestyle factors:** Not applicable to a Mendelian gene defect; heat/exercise exposure matters clinically due to anhidrosis (risk of hyperthermia).
- **Infectious agents:** Not a cause; opportunistic pathogens (CMV, EBV, Pneumocystis jirovecii, candida, other viruses/bacteria/fungi) act as **downstream complications** of the T-cell defect [PMID: 26469693; 26109647].

6. Mechanism / Pathophysiology

Causal chain (initiating lesion → clinical manifestation)

1. **Biallelic LOF mutation in *ORAI1*** → loss/non-function of the CRAC-channel pore subunit in the plasma membrane. [Human clinical/in vitro;

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2. This **results in abolished store-operated Ca^{2+} entry (SOCE)**: after antigen-receptor engagement, $\text{PLC}\gamma \rightarrow \text{IP}_3$ depletes ER $\text{Ca}^{2+} \rightarrow \text{STIM1}$ oligomerizes and senses depletion → but with no functional *ORAI1*, **STIM1 cannot open a Ca^{2+} channel**, so sustained cytosolic Ca^{2+} influx fails. [In vitro;

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3. Loss of the sustained Ca^{2+} signal **leads to failure of calcineurin-mediated NFAT dephosphorylation** → NFAT stays cytoplasmic → transcription of IL-2 and other

cytokine/chemokine genes fails. [In vitro;

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4. In T cells this **results in impaired proliferation, cytokine production, and effector differentiation** despite *normal* lymphocyte numbers → **SCID-like combined immunodeficiency** → recurrent/opportunistic infections. [Human clinical;

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5. **Branch (autoimmunity):** SOCE loss **results in strongly reduced invariant NKT and regulatory T (Treg) cells** and altered $\gamma\delta$ -T/NK subsets → **breakdown of peripheral tolerance** → autoimmune cytopenias, lymphoproliferation, HLH. [Human clinical;

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6. **Branch (ectodermal — sweat glands):** In eccrine sweat-gland epithelium, SOCE drives fluid/electrolyte secretion; ORAI1 loss **leads to defective sweat secretion → anhidrosis → heat intolerance.** [Inferred from tissue expression + phenotype;

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7. **Branch (ectodermal — teeth):** In **ameloblasts/enamel cells**, CRAC-channel (ORAI1/2/3 + STIM1/2) SOCE supplies the large Ca^{2+} flux needed for enamel mineralization — demonstrated directly in primary enamel cells (thapsigargin-evoked SOCE blocked by the CRAC inhibitor Synta-66), most abundant at the maturation stage; ORAI1 loss **leads to hypomineralized enamel → amelogenesis imperfecta.** [Model/in vitro + human;

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8. **Branch (muscle):** SOCE is required for skeletal-muscle Ca^{2+} homeostasis and development; when impaired, muscle is prone to early fatigue and myopathy, so ORAI1 loss **leads to a non-progressive congenital myopathy / muscular hypotonia** (a tissue-autonomous defect

not corrected by HSCT). [Human clinical + model;

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Note the symmetry: *GOF STIM1/ORAI1* → constitutive SOCE → tubular aggregate myopathy

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— both excess and deficiency of SOCE injure muscle.

Category detail

- **Molecular pathways:** Store-operated Ca^{2+} entry / CRAC-channel pathway; Ca^{2+} –calcineurin–NFAT signaling (Reactome R-HSA-2672351 "Stimuli-sensing / STIM-ORAI"; KEGG "Calcium signaling pathway" hsa04020). Downstream NFAT target-gene transcription.
- **Cellular processes:** T-cell activation, proliferation, cytokine secretion; immune synapse Ca^{2+} signaling; Treg/iNKT development; sweat-gland fluid secretion; ameloblast biomineralization; myocyte Ca^{2+} handling.
- **Protein dysfunction:** Loss of function — misfolded/mis-gated (missense) or absent (frameshift/nonsense) ORAI1 → non-conducting or absent CRAC channel. Not aggregation-related.
- **Metabolic changes:** No primary metabolic defect; secondary effects on Ca^{2+} -dependent mitochondrial/energy signaling possible.
- **Immune system involvement:** Central — combined immunodeficiency **plus** autoimmunity (immunodysregulation), a hallmark "immunodeficiency-with-autoimmunity" pattern.
- **Tissue damage mechanisms:** Infection-driven tissue injury; autoimmune destruction of blood cells; HLH-associated inflammation.
- **Biochemical abnormality: Ion-channel defect** — loss of Ca^{2+} -selective CRAC conductance (the defining biochemical lesion).
- **Molecular profiling:** Patient T cells/fibroblasts show **absent SOCE** on Ca^{2+} imaging and **absent/greatly reduced CRAC current (I_{CRAC})** on patch-clamp; reduced ORAI1 protein by Western/flow [PMID: 29155098]. No large transcriptomic/proteomic/metabolomic disease atlases exist (ultra-rare).
- **Functional genomics:** *Orai1* and *Stim1/2* conditional-knockout mice recapitulate SOCE loss and immune defects, confirming causality [Model organism;

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Suggested ontology terms

- **GO (BP):** GO:0002115 (store-operated calcium entry), GO:0070588 (calcium ion transmembrane transport), GO:0033173 (calcineurin-NFAT signaling cascade), GO:0042110 (T cell activation).
 - **GO (MF):** GO:0015279 (store-operated calcium channel activity).
 - **GO (CC):** GO:0005886 (plasma membrane), GO:0034704 (calcium channel complex).
 - **CL:** CL:0000084 (T cell), CL:0000815 (regulatory T cell), CL:0000921 (NKT cell), CL:0002064 (ameloblast/enamel-forming cell), CL:0000188 (skeletal muscle cell).
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7. Anatomical Structures Affected

- **Organ / system level:**
 - **Primary:** Immune system (thymus-derived T-cell compartment, lymphoid tissues) — UBERON:0002405 (immune system); **integument/skin adnexa — eccrine sweat glands** (UBERON:0001820); **teeth/enamel organ** (UBERON:0001091 tooth; UBERON:0007375 enamel); **skeletal muscle** (UBERON:0001134).
 - **Secondary:** Blood (autoimmune cytopenias, UBERON:0000178); spleen/liver/lymph nodes (lymphoproliferation); lungs/GI (site of recurrent infection).
 - **Body systems:** Immune, integumentary, musculoskeletal, hematologic, dental/craniofacial.
 - **Tissue / cell level:**
 - Epithelial: sweat-gland secretory epithelium; ameloblasts (enamel epithelium).
 - Immune: T lymphocytes (esp. Treg, iNKT, $\gamma\delta$ -T), NK cells.
 - Muscle: skeletal myofibers.
 - **Cell Ontology:** CL:0000084, CL:0000815, CL:0000921, CL:0002064, CL:0000188, CL:0000623 (NK cell).
 - **Subcellular level:** **Plasma membrane** (site of the ORAI1/CRAC channel) and **endoplasmic/sarcoplasmic reticulum** (STIM1 sensor). GO-CC: GO:0005886 (plasma membrane), GO:0005783 (endoplasmic reticulum), GO:0034704 (calcium channel complex).
 - **Localization / lateralization:** Systemic and **bilateral/symmetric** (generalized anhidrosis, generalized enamel involvement, generalized hypotonia); not focal or lateralized.
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8. Temporal Development

- **Onset: Congenital / neonatal–infantile.** Immunodeficiency typically presents in the **first months of life** with severe infection; muscular hypotonia and anhidrosis are congenital; enamel defects become apparent as teeth erupt (childhood). Onset pattern: **early, chronic**, with acute infectious/autoimmune exacerbations.
 - **Progression:**
 - **Immunodeficiency:** persistent/lifelong unless corrected by HSCT; punctuated by acute infections and autoimmune flares (HLH can be fulminant,
- P** 28633876
- **Muscular hypotonia: non-progressive** (static).
 - **Anhidrosis & enamel defects: stable/lifelong**, non-progressive.
 - **Course pattern:** Chronic lifelong disease with **episodic** infectious/autoimmune crises superimposed.
 - **Remission:** No spontaneous remission of the underlying channel defect. **Treatment-induced immune remission** follows successful HSCT (immune compartment), but ectodermal/muscle features persist because they are non-hematopoietic.
 - **Critical period: Infancy** is the window of highest mortality and the optimal window for definitive immune therapy (early HSCT before severe infection/organ damage).

9. Inheritance and Population

- **Epidemiology: Ultra-rare.** Fewer than ~30 patients with CRAC channelopathy (ORAI1 + STIM1 LOF combined) reported worldwide; ORAI1-LOF kindreds number a handful. Prevalence/incidence too low to estimate reliably (Orphanet: prevalence <1/1,000,000). No SEER/registry rates.
- **Inheritance: Autosomal recessive** (biallelic LOF). [PMID: 16582901, 29155098]
- **Penetrance: Complete** for the biallelic null genotype (all reported homozygotes affected); heterozygotes **unaffected** clinically though with subclinical partial SOCE reduction (gene-dosage) [PMID: 19075015].
- **Expressivity: Variable** across the tetrad — infection/immune severity and degree of muscle/ectodermal involvement differ between kindreds (possible paralog modifiers).

- **Genetic anticipation:** Not applicable (not a repeat-expansion disorder).
- **Germline mosaicism:** Not reported.
- **Founder effects:** Individual kindreds carry private homozygous alleles, typically in **consanguineous families**; no broad founder allele established.
- **Consanguinity: Major** — most reported cases arise from consanguineous unions (hallmark of rare recessive disease).
- **Carrier frequency:** Not established; biallelic pathogenic genotype essentially absent from population databases.
- **Population demographics:** Reported in multiple ethnicities (consanguineous kindreds from various regions); no strong ethnic predilection beyond consanguinity. **Sex ratio ~1:1** (autosomal). Age distribution skews to **infants/young children** given early onset and historically high mortality.

10. Diagnostics

- **Clinical / laboratory tests:**
- **Immunophenotyping (flow cytometry):** *Normal* total T/B/NK counts (distinguishes from typical SCID) but **reduced iNKT and Treg cells**, altered $\gamma\delta$ -T/NK subsets; poor in-vitro T-cell proliferation to mitogens/antigens [PMID: 29155098]. Immunoglobulins may be normal or with impaired specific antibody responses.
- **Functional Ca^{2+} assay (diagnostic gold standard for the mechanism):** **Absent SOCE** on single-cell Ca^{2+} imaging of patient T cells/fibroblasts; **absent I_{CRAC}** by patch-clamp [In vitro;

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- **ORAI1 protein expression:** reduced/absent by Western blot/flow.
- CBC (autoimmune cytopenias), Coombs test (AIHA), ferritin/triglycerides/sIL-2R (HLH work-up).
- **Sweat testing** (anhidrosis); dental exam (enamel hypoplasia/amelogenesis imperfecta); muscle exam/CK (usually normal-to-mildly abnormal; hypotonia clinical).

- **Biomarkers:** No circulating molecular biomarker; the functional SOCE defect and *ORAI1* genotype are the definitive markers. LOINC codes apply to component tests (lymphocyte subsets, Ig levels).
- **Imaging / electrophysiology:** No pathognomonic imaging. EMG/muscle biopsy may show non-specific/mild myopathic changes; **patch-clamp electrophysiology** (research) demonstrates absent CRAC current.
- **Biopsy/pathology:** Sweat-gland biopsy shows structurally present but non-functional eccrine glands; tooth histology shows hypomineralized enamel.
- **Genetic testing (definitive):**
- **Single-gene sequencing of *ORAI1***, or (preferred first-line today) **primary-immunodeficiency gene panels** or **whole-exome sequencing (WES) / whole-genome sequencing (WGS)** identifying biallelic *ORAI1* LOF variants; confirm recessive segregation in parents [GTR/ClinVar].
- CMA/karyotype/FISH: **not indicated** (no CNVs). mtDNA/repeat-expansion testing: not applicable.
- **Clinical criteria:** No formal consensus criteria; diagnosis rests on the phenotypic tetrad + demonstration of absent SOCE + biallelic *ORAI1* LOF.
- **Differential diagnosis:** Classical **SCID** (usually low T-cell counts — *ORAI1* has normal counts); **STIM1-LOF CRAC channelopathy** (indistinguishable clinically — test *STIM1*); **NEMO/IKBKG (EDA-ID)** and **NFKBIA** anhidrotic ectodermal dysplasia with immunodeficiency (distinct —

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notes *ORAI1* form is distinct); **ALPS** and other IPEX-like immunodysregulation for the autoimmune component; primary HLH genes (*PRF1*, *UNC13D*) when HLH is the presentation

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- **Screening:** No newborn TREC screening detection (T-cell numbers normal, so ***ORAI1* deficiency is typically MISSED by TREC-based newborn SCID screening** — an important caveat). Cascade **carrier testing** in consanguineous families and prenatal/preimplantation testing where the familial variant is known.

11. Outcome / Prognosis

- **Survival / mortality:** Historically **poor without immune reconstitution** — early death from overwhelming infection or fulminant autoimmunity/HLH is common in infancy/early childhood. With **successful HSCT**, immunological prognosis improves substantially.
- **Morbidity / function:** Even after immune cure, patients retain **lifelong anhidrosis (heat intolerance), dental enamel disease, and muscular hypotonia** because these are non-hematopoietic and not corrected by HSCT — a key prognostic point
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- **Complications:** Opportunistic/severe infections; autoimmune hemolytic anemia and thrombocytopenia; lymphoproliferation; **HLH** (can be presenting and life-threatening,

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dental morbidity; hyperthermia episodes.
- **Recovery potential:** Immune compartment can be reconstituted (curative for infection/autoimmunity) via HSCT; non-immune features are permanent.
- **Prognostic factors:** Age at diagnosis, severity/organ damage from infection before treatment, occurrence of HLH, and success/timing of HSCT. No validated molecular prognostic biomarker beyond genotype.

12. Treatment

(No ORAI1-specific approved drug exists; management is that of combined immunodeficiency with immunodysregulation. NCIT terms suggested.)

- **Definitive / curative:**
- **Allogeneic hematopoietic stem cell transplantation (HSCT)** — corrects the hematopoietic/immune defect (donor immune cells have wild-type ORAI1). **Does not** correct sweat-gland, dental, or muscle disease. NCIT: C15431 (Hematopoietic Stem Cell Transplantation). [Supported by disease-mechanism logic + CID management standards;

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- **Pharmacotherapy / supportive (symptom & complication management):**

- **Immunoglobulin replacement (IVIG/SCIG)** for antibody insufficiency — NCIT: C578 (Immunoglobulin Therapy).
 - **Antimicrobial prophylaxis** (e.g., trimethoprim-sulfamethoxazole against *Pneumocystis*; antifungal; antiviral) — NCIT: C15843 (Antibiotic Therapy).
 - **Immunosuppression** for autoimmunity/HLH (corticosteroids, and HLH-directed therapy e.g., etoposide-based protocols when HLH occurs) — NCIT: C15370 (Immunosuppressive Therapy).
 - **Blood product/transfusion support** for autoimmune cytopenias.
 - **Advanced / experimental: Gene therapy / gene correction** of *ORAI1* in autologous HSCs is a conceptual future approach (not clinically available). No approved RNA/targeted/immunotherapy specific to *ORAI1* deficiency. (Note: CRAC-channel *inhibitors*, e.g., under development for autoimmune/inflammatory disease, are the opposite pharmacology and are contraindicated conceptually here.)
 - **Supportive / rehabilitative: Heat-avoidance and cooling strategies** for anhidrosis; **dental restoration/prosthodontics** for enamel defects; **physiotherapy** for hypotonia; nutritional support.
 - **Pharmacogenomics**: Not applicable specifically; standard HSCT-conditioning PGx (e.g., thiopurine/TPMT, busulfan monitoring) applies generically.
 - **Treatment algorithm**: Early diagnosis → infection/autoimmune stabilization + prophylaxis + IVIG → definitive **HSCT** → lifelong management of non-immune (dental/sweat/muscle) features and heat precautions.
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13. Prevention

- **Primary prevention**: No way to prevent the genetic lesion. **Genetic counseling** for consanguineous couples/known carriers; **carrier screening**, **prenatal diagnosis**, and **preimplantation genetic testing (PGT)** where the familial *ORAI1* variant is known.
- **Secondary prevention (early detection/treatment)**: Early clinical suspicion (infant with infections + anhidrosis + hypotonia + enamel defects) → SOCE functional assay + *ORAI1* sequencing → early HSCT. **Caveat**: standard TREC newborn screening does not detect it (normal T-cell counts).

- **Tertiary prevention (complication reduction):** Infection prophylaxis, IVIG, vaccination caution (avoid **live vaccines** in immunodeficient patients), aggressive management of autoimmunity/HLH, heat-precautions, dental care.
 - **Immunization:** Killed/inactivated vaccines per immunodeficiency guidelines; **live vaccines contraindicated**.
 - **Counseling:** Autosomal-recessive counseling — 25% recurrence risk per pregnancy for carrier couples; cascade testing of relatives.
 - **Public-health/environmental:** Not applicable (monogenic disease).
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14. Other Species / Natural Disease

- **Taxonomy / orthologs:** *ORAI1* is conserved across vertebrates. **Mouse *Orai1*** (NCBI Gene ID 109305; MGI:1925542), **rat *Orai1***, **zebrafish**, and a *Drosophila* ortholog (*Orai*) exist — the pathway was in part dissected in *Drosophila* RNAi screens.
 - **Natural disease in other species:** A **naturally occurring ORAI1-associated disorder** in companion/wildlife species is **not well documented** in OMIA; the disease knowledge base is dominated by human patients and engineered animal models. (Not applicable / no strong natural-disease entry.)
 - **Comparative biology:** SOCE and the STIM–ORAI mechanism are **evolutionarily conserved** from insects to mammals, making cross-species mechanistic inference valid. Mouse SOCE loss reproduces immune phenotypes, supporting conserved mechanism.
 - **Zoonotic potential / transmission:** None (genetic, non-communicable).
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15. Model Organisms

- **Mouse (primary model; *Mus musculus*, NCBI Taxon 10090):**
- **Global *Orai1* knockout:** severe SOCE loss; reported **perinatal lethality/reduced viability and small body size** in some backgrounds, with immune defects — recapitulating the essentiality of CRAC channels. Surviving KO/knock-in models show impaired T-cell function.
- ***Orai1* R93W knock-in** (mouse equivalent of human R91W) models the human SCID allele.

- **Conditional/tissue-specific knockouts** (e.g., *Stim1/Stim2* double conditional; muscle-specific *Orai1* KO) dissect cell-type roles — e.g., muscle-specific *Orai1* deletion abolishes muscle SOCE

([P 35939054](#)),

and *Stim1/Stim2*-deleted macrophages/DCs surprisingly retain many effector functions, showing lineage-specific SOCE dependence

([P 26109647](#)).

- **Model types available:** knockout, knock-in (point mutant), conditional/floxed, tissue-specific — via MGI/IMPC/KOMP.

- **In vitro / cellular models: Patient-derived T cells and fibroblasts** (absent SOCE/I_CRAC) are the definitive human cellular models [PMID: 29155098, 16582901]; heterologous ORAI1/STIM1 expression systems (HEK293) and **concatenated tetramer constructs** quantify gene-dosage effects of R91W [PMID: 19075015]. Ameloblast/enamel-cell models illustrate CRAC-dependent mineralization

([P 30114531](#)).

- **Invertebrate:** *Drosophila* Orai/Stim genetic screens established the pathway.
- **Phenotype recapitulation:** Mouse and patient-cell models faithfully reproduce the **SOCE/immune defect**; mouse models capture immune and muscle phenotypes well.
- **Model limitations:** Complete global *Orai1* KO can be perinatally lethal in mice (more severe than the human hypomorphic/viable phenotype), and murine ectodermal (sweat/enamel) and autoimmune features may not fully phenocopy the human tetrad; paralog compensation differs between species.
- **Resources:** MGI (*Orai1* MGI:1925542), IMPC/KOMP, IMSR for strain availability; ZFIN (zebrafish), FlyBase (*Drosophila Orai*).

Supported vs Refuted Hypotheses

- **SUPPORTED:** ORAI1 LOF → abolished SOCE → NFAT-signaling failure → combined immunodeficiency

([P 16582901](#))

(P 17572487

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(P 26469693

✓

- **SUPPORTED:** ORAI1 loss reduces iNKT/Treg cells → autoimmunity/immunodysregulation

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P 29155098

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P 28633876

✓

- **SUPPORTED:** Multisystem (sweat gland, dental enamel, muscle) involvement reflects SOCE dependence of those tissues

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P 30114531

✓

- **SUPPORTED:** Recessive with dosage-dependent channel function (heterozygotes subclinical)

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P 19075015

✓

- **CONTRAST/REFUTED as same disease:** Gain-of-function *ORAI1* is NOT "ORAI1 deficiency"; it causes dominant TAM/Stormorken

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✓ (kept distinct)

Limitations & Future Directions

- Ultra-rare disease: all clinical claims rest on small case series; no registry-level epidemiology, QoL, or survival statistics exist. Prevalence, penetrance-by-feature, and long-term HSCT outcomes are under-quantified.
- Some branch mechanisms (sweat gland, enamel) are **inferred** from tissue SOCE dependence and mouse data rather than proven in human tissue in vivo.

- Future: prospective natural-history registries; gene-correction/gene-therapy proof-of-concept; therapies for the non-hematopoietic (dental/sweat/muscle) features not addressed by HSCT.

Evidence key PMIDs: 16582901 (Feske 2006, first ORAI1 SCID mutation), 29155098 (novel recessive ORAI1 mutations, subset deficits), 26469693 (Lacruz & Feske review, disease definition & GOF contrast), 17572487 (SOCE–NFAT axis), 19075015 (heterozygous gene-dosage), 28633876 (HLH presentation), 30114531 (enamel/CRAC), 26109647 (innate-cell SOCE dependence, mouse), 35939054 (muscle-specific Orai1 KO), 29635109 (ion channelopathies of immune system review).

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